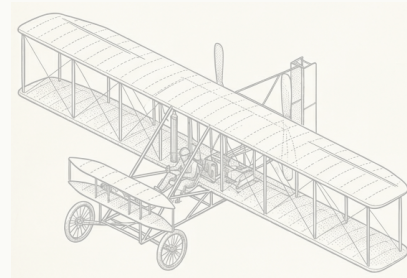


1903

FIG. 01



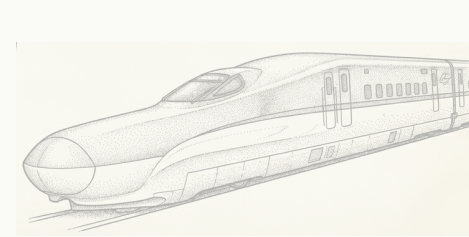
BICYCLE



FLYING MACHINE

1997

FIG. 02



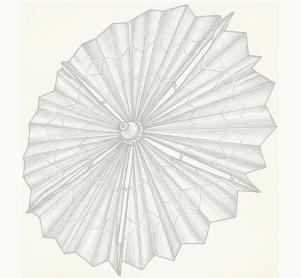
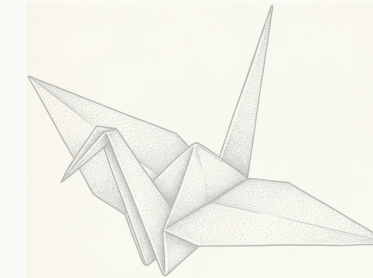
KINGFISHER BEAK



SHINKANSEN NOSE

2016

FIG. 03



ORIGAMI



SOLAR SAIL

Carnegie
Mellon
University



Human-
Computer
Interaction
Institute

KITTUR LAB · HUMAN-COMPUTER INTERACTION INSTITUTE

Solving industry's hardest problems with solutions from *unexpected places*.

What the research offers, which companies to bring us, and how it can help you open doors.

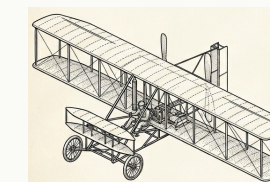
THE PROBLEM EVERY COMPANY BRINGS

Every R&D team is stuck on problems they're making only *incremental progress* on.

Because experts search where they already know — and AI gets you more of what's already out there.

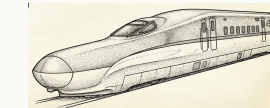
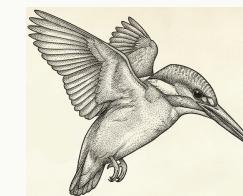
Our research makes finding breakthroughs from distant domains *systematic*: retrieve by mechanism, not keyword; map it into their domain; de-risk it into something their team can build and test.

BREAKTHROUGHS COME FROM ANALOGIES



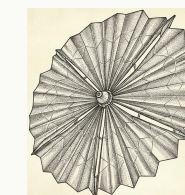
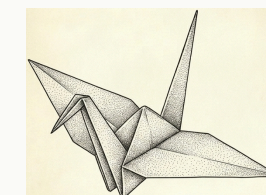
Bicycle → Wright Flyer (1903)

Frame-lean steering became wing-warp steering: the control problem that grounded every prior aviator.



Kingfisher → bullet train (1997)

A beak that enters water without a splash became a nose that ends sonic booms in tunnels.



Origami → NASA solar sail (2016)

A Miura fold packs a 25 m sail into a 1 m fairing: a packaging problem became a paper-folding one.

Serendipity found these. We make them searchable — forward, and in reverse (a capability → the problems it could solve).

12+ years of research at HCII PNAS · best papers at CHI, CSCW, KDD

NSF-funded · 4 patents Kittur · Martelaro · Mohanty · Shahaf

WHAT IT LOOKS LIKE · 60 SECONDS

A public P&G challenge: *clean between teeth without floss.*

Nobody in oral care would look at an oil rig. The engine did – and came back with a concept, its evidence, and the tests to run.



01 SEARCH

Retrieve by *mechanism*: “dislodge a stubborn deposit from a narrow channel.” Matches came from **drilling fluids** that scour a borehole and **vascular medicine**, where magnetic bead swarms clear clots.

02 TRANSFER

Map the mechanism onto the real constraints: safe in the mouth, no hardware a consumer won’t use, a sachet and a rinse. Every claim is tied to a primary source – not a hallucinated one.

03 EVALUATE

Hand back the four test protocols and success criteria that settle the risk: chain formation, biofilm softening, debris removal, bead recovery. The team knows what to run Monday.

The same loop runs on a materials scientist’s holy-grail problem or a propulsion chemist’s storage problem – with their experts in the loop.

PROOF · WITH INDUSTRY PARTNERS

Ideas their own experts rated as ones *they couldn't have reached.*

COVESTRO · FORTUNE 500 MATERIALS

3 “HOLY-GRAIL” PROBLEMS STUCK 10+ YEARS · 17 SCIENTISTS

54%

more novel, no drop in feasibility ($p < .05$)

30+

promising ideas vs. 0–1 from their own LLM tools

2

already in their managed innovation pipeline

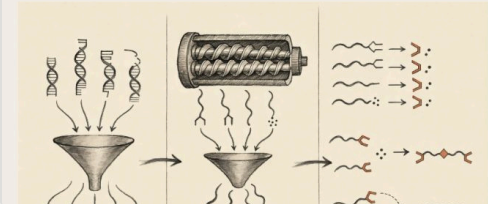
IDEA 01 OF 03

NEXT >

BORROWED FROM DNA-SEQUENCING PREPARATION

Convergent chain repair: boron funnels every broken end into one joinable handle

Borrow the terminus-funneling step from next-generation sequencing library preparation to converge every broken polyolefin chain end onto one boron handle and rebuild linear molecular weight without radicals or gel.



HOW IT WORKS

Thexylborane, carrying exactly two B–H bonds, hydroborates terminal vinyl and vinylidene chain ends in the melt, capturing two chains per molecule onto one dialkylborane handle; a solid-acid co-additive funnels carbonyl-derived ends into the same pathway via dehydration. Downstream CO injection drives Brown carbonylation ③, migrating both chains onto a carbonyl carbon to yield a linear ketone bridge – purely ionic, no radical generated, mid-chain attack impossible. No one has combined

“If we sat down and spent a week going through these things, that would be a worthwhile investment to make sure that we’re spending the next two years on the right projects.” — R&D group leader

URSA MAJOR · AEROSPACE PROPULSION “HOW CAN WE EXTEND THE LIFE OF OUR PROPELLANT?”

+80%

estimated propellant life at minimal extra cost vs. 5–10% from their previous approach

Paid pilot underway.

Concepts borrowed from semiconductor wafer gettering and nuclear-waste ceramics; scope expanding.

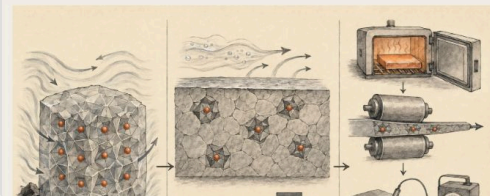
IDEA 03 OF 03

< PREV

BORROWED FROM NUCLEAR-WASTE CERAMICS

Lattice-pinned alloy: heat treatment locks trace metals where peroxide cannot reach

Borrow lattice-site immobilization from nuclear-waste ceramics to trap catalytic trace metals in aluminum tank plate into crystal sites hydrogen peroxide physically cannot breach, enabling decade-scale storage without hazardous passivation.



HOW IT WORKS

Annealing 5254 plate at 480 °C drives residual Fe and Cu into Al₃Fe and Al₆(Fe,Mn) intermetallics ②⑤, ③② whose formation enthalpies exceed the oxidation potential H₂O₂ can deliver, locking catalytic metals in lattice sites the oxidizer cannot breach. Current practice controls only composition and surface chemistry ③④; this opens a third axis – microstructural site occupancy – set by a single upstream heat treatment before cold-rolling to flight strength. Intermetallics formed above 400 °C are

“If this was [available to every engineer] they would [use it] daily.” “...ideas which no one in my team could come up with.” — R&D leader, Fortune 500 automotive

BUILT ALONGSIDE Covestro · Ursa Major · Toyota · Bosch · Conservation X Labs · and a public P&G challenge

WHO TO BRING US

Industrial R&D with a *named, stuck problem* – or an underused capability.

partner in conversation plain: target

MATERIALS & CHEMICALS

Covestro	BASF	PPG	Dow	DuPont
3M	Henkel	Solvay	Evonik	SABIC
Arkema				

AUTOMOTIVE & MOBILITY

Toyota	Bosch	GM	Ford	Stellantis
Honda	Magna	Cummins		

CONSUMER GOODS & FOOD

P&G	PepsiCo	Chobani	Unilever	Nestlé
Mondelez	Mars	L'Oréal	Kraft Heinz	
Colgate-Palmolive				

ENERGY & INDUSTRIAL

Westinghouse	Wabtec	Alcoa	GE Vernova
Siemens Energy	SLB	Eaton	Caterpillar

AEROSPACE & DEFENSE

Ursa Major	RTX	Lockheed Martin
Northrop Grumman	Boeing	Honeywell
Anduril	Blue Origin	

ELECTRONICS & SEMICONDUCTORS

Mitsubishi Electric	Intel	Micron	Corning
Applied Materials	Samsung	TSMC	

HOW TO SPOT A FIT

The right person, the signal, and *the question that opens it.*

01 THE RIGHT PERSON

VP or Head of R&D

CTO · Chief Science or Chief Innovation Officer

Open-innovation and alliance-management leads

The BD lead looking for new markets for an existing material or IP

Not the CIO, and not procurement — the person who owns the technical roadmap and the problems on it.

02 THE SIGNAL IT'S A FIT

A problem their team has been on for years

A paid-for capability whose best markets they can't see

They've tried the LLMs and been underwhelmed

An open-innovation mandate and appetite for a pilot

One of these is enough. Two and it's a strong lead.

03 THE QUESTION THAT OPENS IT

“What’s the problem your best people have been on the longest?”

“What have you paid to develop that you suspect has uses you can’t see from here?”

“Where did the AI tools your teams tried fall short?”

Send us the answer. We come back with directions and the CMU people who could pursue them.

HOW THIS CAN HELP YOU

Walk into the first meeting with **three concrete directions** — and the CMU people who could pursue them.

01 Start the conversation

A company's interest is nebulous, or they have a capability and no idea where it fits. Send it to us. We run it through the engine and hand you back concrete directions and the faculty across CMU who could pursue each one — a reason to meet, not a brochure.

For production deployment inside a company there is also a CMU spinout, Analogical Engines, built on this research — so an inquiry can be routed to the right door: exploratory work through the lab, deployment through the company.

02 Sponsored research & workshops

Through the lab: a half-day ideation session with their R&D team on a named problem, or a scoped sponsored project on the thing they've been stuck on. Their experts stay in the loop; our students and postdocs do the work.

03 Matchmaking across CMU

Run it in reverse: from a company's capability to the problems across campus it could solve. It makes your outreach strategic — the intro comes with a match, not just to us.

BRING ME A COMPANY AND A PROBLEM

nkittur.github.io/unexpected-places

Case studies, what we offer, who to bring · nkittur@cs.cmu.edu

Happy to talk to any of you.

Aniket Kittur